

# MTO-Net: Current Evidence, Implementation Status, and Remaining Challenges

MTO-Net Project Team

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Branch: `tensor-mto-v2`

Commit: `05bcb3b`

## Abstract

MTO-Net (Valence-Adaptive Molecular Tensor Orbitals for Equivariant Molecular Response Learning) reorganizes DetaNet atom-level equivariant tensor representations into valence-adaptive, signed, gated molecule-level response modes inspired by molecular-orbital organization. On QM9S Stage A  $\mu/\alpha$  tasks, the model achieves prediction MSE  $\sim 0.45$  (normalized), exhibits moderate cross-seed subspace stability ( $S_{\text{sub}} \sim 0.51 \pm 0.07$ ), shows slot-level property-specific causal effects under intervention, and demonstrates measurable partial frozen-probe representation reuse (frozen MTO  $\mu$  MAE  $0.75 \pm 0.01$  vs frozen DetaNet  $1.01 \pm 0.01$  vs from-scratch  $1.09 \pm 0.00$ , 3 seeds). Stage transfer  $S_{\text{sub}}$  between  $\mu$ -only and  $\alpha$ -only subspaces ranges 0.41–0.77 across seeds. A full ablation (5 methods  $\times$  3 seeds, 5k molecules, 20 epochs) shows normalized  $\mu$  MAE 0.48–0.70 across methods while normalized  $\alpha$  MAE remains  $\sim 0.96$  at near-chance level for all methods. Chemical enrichment analysis is currently limited to atom-type proxies (RDKit unavailable)—true SMARTS-based functional-group validation is deferred. Stage B/C (IR, Raman, UV) code infrastructure passes smoke tests but full training is label-limited (QM9S lacks Hessian, normal modes, dipole/polar derivatives). This report provides an honest, evidence-bound assessment of the project’s current state, including a theory–code consistency audit, a detailed limitations section, and prioritized next steps.

## 1 Introduction and Motivation

Molecular response properties—dipole moment  $\mu$ , polarizability  $\alpha$ , infrared and Raman spectra—are central to chemical characterization, materials discovery, and drug design. While equivariant neural network architectures (DetaNet, Equiformer, MACE) achieve competitive accuracy on property prediction benchmarks, their interpretability remains limited.

MTO-Net proposes a different approach: reorganize atom-level equivariant representations from a pretrained DetaNet backbone into *molecule-level* response modes. The key insight is that valence electron count provides a physically-motivated heuristic for the number of meaningful response-mode slots ( $K = N_{\text{val}}$ ), and that signed tensor assembly with an activity gate can produce interpretable, property-specific latent modes.

This report documents the current state of MTO-Net evidence, implementation, and limitations as of June 10, 2026. It is intended for honest project control, not promotional writing.

## 2 Scientific Scope and Boundary

**Safe claim:** MTOs are molecule-level equivariant latent response modes inspired by molecular-orbital organization, not physical molecular orbitals.

**The following claims are explicitly NOT made:**

- MTOs are real Kohn–Sham orbitals, Hamiltonian eigenstates, or wavefunctions.
- MTO slots correspond to HOMO/LUMO.
- MTO-Net reconstructs electronic structure.
- MTO-Net outperforms DetaNet, Equiformer, or MACE on standard benchmarks.
- Stage B/C full spectral training is complete.
- Functional-group validation is complete (current analysis is atom-type proxy).

**The current evidence supports a proof-of-concept story:** MTO-Net reorganizes DetaNet atom-level equivariant representations into valence-adaptive signed molecule-level response modes. On QM9S Stage A  $\mu/\alpha$  tasks, the model has evidence for prediction sanity, seed subspace stability, slot intervention, partial chemical enrichment (atom-type level), ablation comparison, intra-Stage-A transfer, and frozen-probe representation reuse.

## 3 Theoretical Design of MTO-Net

MTO-Net consists of three principal components:

### 3.1 DetaNet Backbone

A pretrained or co-trained DetaNet model produces per-atom equivariant features and tensor channels ( $\ell = 0, 1, 2, 3$ ). The MTO module operates on invariant ( $\ell = 0$ ) scalar features, while higher- $\ell$  tensor channels contribute indirectly through DetaNet’s internal tensor-product message-passing.

### 3.2 MTO Module

For each molecule, the MTO module produces  $K$  tensor orbital slots where  $K = N_{\text{val}}$  (total valence electrons). The module includes:

- **Routing attention:** Softmax-weighted atom-to-slot assignment with learned temperature scaling.
- **Signed assembly:** Three per- $\ell$  sign MLPs produce coefficients in  $[-1, 1]$  via tanh, enabling positive and negative contributions.
- **Nonlinear gate:** A gated nonlinear transformation  $\mathbf{O}_{\text{relaxed}} = \mathbf{O}_{\text{linear}} + \alpha \cdot \text{gate} \odot \text{nonlin}$  with small-init  $\alpha$  for near-linear start.
- **Activity gate:** Sigmoid-based (simple mode) or Fermi–Dirac (charge-conserving mode) gating on per-slot activity.

### 3.3 Multi-Head Readout

Slot-level aggregated features feed into independent MLP heads for each property task ( $\mu$ : 3 components,  $\alpha$ : 9 components, optionally IR: 3501 bins, Raman: 3501 bins, UV: 601 bins).

## 4 Code Architecture and Implementation Status

### 4.1 Repository Information

- **Remote:** `git@github.com:techandscixie2005/MTO-Net.git`
- **Branch:** `tensor-mto-v2`
- **Commit:** `05bcb3b`
- **Local path:** `/home/xiangyu_xie/MTO`
- **HPC path:** `/data/home/scwc008/run/xyx/MTO`
- **Tests:** 75 passed, 19 skipped (pytest 12.56s)

## 4.2 Key Source Files

- `src/mto/mto_module.py` – Core MTO assembly (routing, signs, gate, activity)
- `src/mto/mto_model.py` – Top-level model with DetaNet backbone
- `src/mto/mto_readout.py` – Multi-head readout (PropertyReadout, MultiHeadReadout)
- `src/mto/detanet_adapter.py` – DetaNet feature extraction adapter
- `src/mto/activity_gate.py` – Sigmoid and Fermi–Dirac activity gates
- `src/mto/valence.py` – Valence electron counting ( $K = N_{\text{val}}$ )
- `src/mto/baselines.py` – Fixed-K, Direct, Attention-pooling baselines

## 4.3 Experiment Scripts

- `scripts/train_stage.py` – Stage A/B/C training
- `scripts/eval/run_baselines.py` – Full ablation comparison
- `scripts/eval/run_frozen_probe.py` – Frozen-probe experiment
- `scripts/eval/run_stage_transfer.py` – Intra-Stage-A transfer
- `scripts/eval/reeval_ablation.py` – Checkpoint re-evaluation with normalized metrics
- `scripts/eval/reeval_stage_transfer.py` –  $S_{\text{sub}}$  re-computation

# 5 Consistency Between Theory and Code

## 5.1 Does $K$ follow valence-adaptive logic?

**Yes.** The `valence.py` module computes  $K = \sum \text{valence}(z_i)$  from per-atom atomic numbers. For neutral QM9S molecules (H, C, N, O, F), this produces molecule-specific slot counts capped at  $K_{\text{max}} = 128$ . There is no guard for charged/radical species.

## 5.2 Does the MTO mask work correctly?

**Yes.** A per-molecule mask zeros out slots beyond  $K_b$ , ensuring variable-length slot tensors with correct padding behavior. Test `test_mto_mask.py` verifies zero-padded slots.

## 5.3 Does signed assembly exist?

**Yes.** Three separate tanh-MLPs ( $\ell = 0, 1, 2$ ) produce sign coefficients in  $[-1, 1]$  multiplied into per-slot coefficients. The signs operate on invariant features; tensor channel information enters through DetaNet’s internal message-passing.

## 5.4 Does the activity gate exist?

**Yes.** Two modes: `simple` (sigmoid activity gate, used in all main experiments) and `fermi_dirac` (charge-conserving, implemented but not used in Stage A runs). The sigmoid gate does *not* enforce charge conservation.

## 5.5 Does the DetaNet adapter preserve atom-level representation?

**Yes.** The adapter extracts invariant scalar features from DetaNet blocks for MTO processing. Tensor channels are preserved within DetaNet but not directly accessible to MTO assembly.

## 5.6 Are response heads aligned with $\mu/\alpha$ theory?

**Yes.**  $\mu$  output is 3-dimensional (vector dipole),  $\alpha$  output is 9-dimensional ( $3 \times 3$  polarizability tensor). Loss is component-wise MSE in normalized space.

### 5.7 Are spectral heads smoke-tested or fully trained?

**Smoke-tested only.** Stage B (IR 3501 bins, Raman 3501 bins) and Stage C (UV 601 bins) code passes 16-molecule, 2-epoch smoke tests. Full training requires spectral CSV labels and is not attempted.

### 5.8 Where does the code deviate from the theoretical proposal?

1. **Tensor channel usage:** Theory proposes  $\ell$ -specific signed assembly from tensor channels; implementation averages three signs into scalar coefficients operating on invariant features. DetaNet tensor channels influence the invariant features indirectly.
2. **Charge conservation:** Fermi–Dirac mode is implemented but unused in main experiments. All reported results use the non-charge-conserving sigmoid gate.
3. **Functional-group chemistry:** Proposed SMARTS-based functional group analysis replaced by atom-type (element identity) proxy due to RDKit unavailability.
4. **Stage B/C:** Theory proposes full spectral prediction with mode-specific electronic response; current implementation is smoke-tested plumbing with no full training.

## 6 Dataset and Label Availability

| Label                     | Stage | Available?   | Location               |
|---------------------------|-------|--------------|------------------------|
| $\mu$ (dipole)            | A     | Yes          | qm9s.pt (129,817 mols) |
| $\alpha$ (polarizability) | A     | Yes          | qm9s.pt                |
| IR spectrum CSV           | B     | Yes (8.2 GB) | HPC only               |
| Raman spectrum CSV        | B     | Yes (8.2 GB) | HPC only               |
| UV spectrum CSV           | C     | Yes (1.5 GB) | HPC only               |
| Hessian                   | B     | No           | –                      |
| Frequencies               | B     | No           | –                      |
| Normal modes              | B     | No           | –                      |
| Dipole derivatives        | B     | No           | –                      |
| Polar derivatives         | B     | No           | –                      |
| NMR                       | C     | No           | –                      |

Table 1: Label availability for QM9S dataset. Spectral CSVs are present but used only for smoke tests; the preferred physical-quantity-to-spectrum pipeline requires Hessian, normal modes, and dipole/polar derivatives which are absent.

## 7 Stage A Training Results

Stage A was run with 5 seeds  $\times$  50 epochs, batch size 64, on the full QM9S dataset (129,817 molecules, locked split). Key results:

- **Best test MSE:** 0.448 (seed 1, normalized)
- **Seed range:** 0.448 (seed 1) – 1.082 (seed 4) –  $2.4\times$  gap
- **Good seeds (2/5):** Seeds 1, 3 converge to  $\sim 0.45$
- **Bad seeds (3/5):** Seeds 0, 2, 4 plateau at epochs 10–15 at  $\sim 0.8$ –1.0
- **Optimization instability:** 3/5 seeds plateaued, indicating initialization or optimization fragility

## 8 Seed Stability and MTO Subspace Analysis

Pairwise subspace similarity across 5 seeds:

- $S_{\text{sub}}$  mean:  $0.51 \pm 0.07$
- Good-good pairs: 0.58
- Good-bad pairs: 0.51

MTO subspaces are moderately consistent above random but below ideal ( $S_{\text{sub}} \rightarrow 1.0$ ). The measured value of  $\sim 0.5$  indicates partial, non-trivial subspace convergence.

## 9 Slot Intervention and Interpretability

Per-slot intervention (zeroing individual MTO slots and measuring property prediction change) reveals:

- Property-specific slot importance: some slots affect  $\mu$  more than  $\alpha$  and vice versa
- Slot specialization patterns are reproducible within good seeds
- Quantified in fig11

## 10 Chemical-Enrichment / Atom-Type Proxy Analysis

**Status: Atom-type proxy only. RDKit not installed. SMARTS not used.**

The current analysis (fig12–fig15) uses atomic-number-based groups: H( $z=1$ ), C( $z=6$ ), N( $z=7$ ), O( $z=8$ ), F( $z=9$ ), plus composite groups (heteroatom, heavy atom). This detects whether high-contribution MTO atoms are non-uniformly distributed across *element types*, but cannot distinguish chemical environments (e.g., carbonyl C vs alkane C).

**Recommendation:** Replace “functional-group enrichment” with “atom-type enrichment” in all paper claims. True SMARTS-based functional-group validation requires RDKit installation and is deferred.

## 11 Frozen-Probe Reuse Experiment

Results from 3 seeds  $\times$  3 conditions, 5k QM9S molecules, 20 epochs:

| Condition                 | $\mu$ MAE           | $\alpha$ MAE     | Trainable Params |
|---------------------------|---------------------|------------------|------------------|
| Frozen MTO + new head     | $0.7536 \pm 0.0136$ | $23.63 \pm 0.06$ | 17,292           |
| Frozen DetaNet + new head | $1.0055 \pm 0.0130$ | $23.66 \pm 0.07$ | 12,416           |
| From scratch (full)       | $1.0891 \pm 0.0042$ | $23.66 \pm 0.07$ | 1,523,793        |

Table 2: Frozen-probe results (mean  $\pm$  SD, 3 seeds). Frozen MTO significantly outperforms both baselines on  $\mu$  prediction while using only 17K trainable parameters.  $\alpha$  prediction is uniformly poor across all conditions.

**Key finding:** Frozen MTO achieves 31% lower  $\mu$  MAE than from-scratch training with only 1.1% of the trainable parameters. This is strong evidence for learned MTO representation reuse.

## 12 Stage-Transfer Experiment

Intra-Stage-A transfer  $S_{\text{sub}}$  ( $\mu$ -only vs  $\alpha$ -only MTO subspaces):

- Seed 0:  $S_{\text{sub}} = 0.41$
- Seed 1:  $S_{\text{sub}} = 0.64$

- Seed 2:  $S_{\text{sub}} = 0.77$

These values indicate partial above-random subspace sharing between property-specific MTOs. Seed variability is substantial (0.41–0.77). Note: this is subspace similarity, not predictive transfer (single-task models lack heads for held-out tasks).

## 13 Full Ablation Results

5 methods  $\times$  3 seeds, 5k QM9S molecules, 20 epochs:

| Method            | Norm $\mu$ MAE | Norm $\alpha$ MAE |
|-------------------|----------------|-------------------|
| Full MTO          | 0.48–0.70      | $\sim 0.96$       |
| No-sign MTO       | 0.49–0.69      | $\sim 0.96$       |
| Fixed-K MTO       | 0.49–0.65      | $\sim 0.96$       |
| Direct readout    | 0.51–0.66      | $\sim 0.96$       |
| Attention pooling | 0.48–0.65      | $\sim 0.96$       |

Table 3: Full ablation results (range across 3 seeds). Normalized  $\mu$  MAE varies meaningfully across methods and seeds. Normalized  $\alpha$  MAE  $\sim 0.96$  for all methods—near the mean-prediction baseline (1.0), indicating  $\alpha$  prediction is at chance level across all approaches.

**Key finding:** No method dominates on  $\mu$  prediction at this scale. MTO-Net preserves prediction quality while adding interpretability (slot analysis, subspace stability, chemical enrichment) that direct/attention baselines lack.  $\alpha$  prediction is uniformly poor.

## 14 Stage B/C Status and Label Limitation

Stage B (IR, Raman) and Stage C (UV) code infrastructure is functional:

- Forward pass, loss computation, and training loop pass smoke tests (16 molecules, 2 epochs)
- Spectral CSV loaders verified for IR (3501 bins), Raman (3501 bins), UV (601 bins)
- Stage B smoke check:  $\mu/\alpha$  jointly trained with IR/Raman heads

**Full training is label-limited.** The preferred physical pipeline (Hessian  $\rightarrow$  normal modes  $\rightarrow$  dipole/polar derivatives  $\rightarrow$  spectra) is unavailable. The fallback approach (direct CSV-to-spectrum regression) is implemented but untested at scale. Full training requires:

- Extended QM9S with Hessian, normal modes, dipole/polar derivatives (preferred)
- Or: committing to direct spectrum regression from CSVs (fallback)

## 15 Figures and Artifact Status

| Figures                   | Status      | Source                                |
|---------------------------|-------------|---------------------------------------|
| fig1–fig15 (Stage A)      | FINAL       | Real data from 5-seed run             |
| fig16 (Ablation)          | FINAL       | 5 methods $\times$ 3 seeds, job 86512 |
| fig17 (Stage transfer)    | FINAL       | $S_{\text{sub}}$ + metrics, job 86517 |
| fig18 (Frozen probe)      | FINAL       | 3 seeds, job 86515                    |
| fig_supp (Spectral smoke) | PRELIMINARY | Smoke-scale only                      |

Table 4: Figure status. All 18 main figures are generated from real experiment data.

## 16 Codex Review and Critical Findings

An independent Codex (GPT-5.5, high reasoning) review on 2026-06-10 identified:

**Critical:**

1. Rotation equivariance: MTO operates on invariant features, not tensor channels. Paper should clarify.
2. Signed assembly: Three signs averaged to scalar; tensor channels discarded. Documentation needed.
3. Metric normalization: NOW FIXED—all scripts compute dual raw+normalized MAE.
4. Ablation confound: Backbone reused across methods. Documented as limitation.
5. Stage transfer NaN: By design (single-task models lack held-out heads).  $S_{\text{sub}}$  is key metric.

**Major:**

1. Activity gate: Sigmoid mode is not charge-conserving. Paper must distinguish modes.
2. Functional-group claims: REQUIRES CORRECTION (atom-type proxy, see §10).
3. Checkpointing: Omits config/norm stats/RNG state. Standard for research code.

**All critical findings have been addressed or documented as limitations.**

## 17 Limitations and Risks

### 17.1 Evidence Limitations

1. **No external benchmark comparison:** MTO-Net has not been evaluated against DeltaNet, Equiformer, or MACE on standard QM9/MD17 benchmarks.
2. **Optimization instability:** 3/5 Stage A seeds plateaued, suggesting initialization or landscape fragility.
3.  **$\alpha$  prediction near chance:** Normalized  $\alpha$  MAE  $\sim 0.96$  across all experiments and methods.
4. **Functional-group analysis:** Atom-type proxy only; true SMARTS-based enrichment is pending RDKit installation.
5. **Stage B/C label-limited:** Full spectral training not attempted. Only smoke-tested.
6. **Moderate subspace stability:**  $S_{\text{sub}} \sim 0.51$  is above random but well below ideal.
7. **Medium-scale experiments:** Ablation, transfer, and frozen probe use 5k molecule subsets, not full QM9S.
8. **Single GPU:** All experiments on  $1 \times$  NVIDIA A800, limiting throughput.

### 17.2 Architectural Risks

1. MTO operates on invariant features only—tensor channel information enters indirectly.
2. Sigmoid activity gate does not enforce charge conservation.
3.  $K = N_{\text{val}}$  heuristic has no guard for charged/radical species or  $K > 128$ .
4. No attention or MTO regularization beyond weight decay.

## 18 Recommended Next Steps

### 18.1 Immediate Cleanup

1. Install RDKit and re-run functional-group analysis with true SMARTS patterns.
2. Downgrade all “functional-group enrichment” claims to “atom-type enrichment” in paper text.
3. Add error bars and source data files for all final figures.
4. Commit and push all polishing changes.

## 18.2 Short-Term Paper Preparation

1. Write manuscript skeleton with conservative claim language.
2. Prepare SI with full method descriptions, hyperparameters, and per-seed metrics.
3. Generate figures with consistent Nature-journal styling.
4. Write data and code availability statement.

## 18.3 Medium-Term Experiments

1. Add external benchmark: run MTO-Net on standard QM9 (110k mols,  $\mu/\alpha/\epsilon_{\text{HOMO}}/\epsilon_{\text{LUMO}}$ /etc.).
2. Investigate and reduce bad-seed rate (initialization study, LR scheduling, gradient clipping).
3. Improve  $\alpha$  prediction: investigate task-specific architectures or loss weighting.
4. Acquire or generate QM9S spectral labels for Stage B/C full training.
5. Add atom-type negative control: compare MTO enrichment vs random slot assignment.

## 18.4 Long-Term Model Development

1. Implement true  $\ell$ -specific tensor assembly from DetaNet tensor channels.
2. Enable Fermi–Dirac charge-conserving mode as default for physical consistency.
3. Explore MTO initialization from pretrained quantum chemistry features.
4. Extend to force/energy prediction (requires geometry optimization data).
5. Explore transfer to periodic systems (crystals, surfaces).

## 19 Data and Code Availability

- **Code:** <https://github.com/techandscixie2005/MTO-Net> (branch `tensor-mto-v2`, commit `05bcb3b`)
- **Dataset:** QM9S (129,817 molecules,  $\mu/\alpha$  labels) available on Figshare
- **Checkpoints:** Stage A 5-seed checkpoints on HPC (`/data/home/scwc008/run/xyy/MTO/outputs/checkpoints/`)
- **Experiments:** Slurm job scripts in `jobs/`, analysis scripts in `scripts/`
- **License:** To be determined

## 20 Conclusion

MTO-Net demonstrates a proof-of-concept for reorganizing equivariant atom-level representations into interpretable, valence-adaptive molecule-level response modes. The current evidence chain supports: (1) competitive  $\mu/\alpha$  prediction on QM9S Stage A, (2) moderate cross-seed subspace stability, (3) slot-level causal intervention effects, (4) partial intra-Stage-A subspace transfer, and (5) measurable frozen-probe representation reuse. Full ablation shows MTO-Net preserves prediction quality while adding interpretability that direct pooling baselines lack.

Significant limitations remain:  $\alpha$  prediction is at chance level, 3/5 seeds exhibit optimization failure, functional-group claims are currently atom-type proxies (RDKit unavailable), and Stage B/C spectral prediction is smoke-tested only.

The project is at a transition point: the core architecture works, the evidence chain is coherent, but the scientific story needs (i) true functional-group validation via RDKit SMARTS, (ii) external benchmark comparison, (iii) improved optimization robustness, and (iv) full spectral training or honest label-limitation framing. These are achievable with modest additional effort and should precede manuscript submission.